

KOTA KINABALU INTL, Malaysia

WMO: 964710

Lat: 5.937N

Lon: 116.051E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.1	73.5	67.8	102.1	76.9	69.1	106.8	77.1	13.1	81.8	11.8	82.1	5.3	100	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.3	93.4	82.9	92.6	82.5	91.6	81.9	84.3	92.0	83.5	91.2	82.8	90.5	4.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.3	168.2	91.2	81.0	161.1	90.0	80.7	159.3	89.7	48.5	91.5	47.5	91.9	46.9	90.4	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.3	9.6	DB	70.4	95.3	2.1	1.6	68.9	96.4	67.7	97.3	66.5	98.2	65.0	99.3	(4)
(5)			WB	67.2	86.0	2.1	1.6	65.7	87.1	64.5	88.1	63.4	88.9	61.9	90.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	81.3	81.3	82.3	83.6	83.9	83.2	82.8	82.9	82.6	82.2	82.2	82.0	(6)	
	DBStd	1.78	1.61	1.54	1.53	1.43	1.72	1.74	1.68	1.74	1.55	1.64	1.39	1.56	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11871	969	876	1001	1007	1051	997	1015	1019	978	999	967	992	(10)	
	CDD65	6396	504	456	536	557	586	547	550	554	528	534	517	527	(11)	
	CDH74	66542	4775	4413	5574	6227	6615	5917	5859	5990	5530	5327	5127	5188	(12)	
	CDH80	25421	1573	1469	2063	2568	2784	2382	2350	2427	2153	1963	1853	1836	(13)	
(14) Wind	WSAvg	4.6	4.2	4.1	4.1	4.4	4.6	4.8	4.9	5.2	5.0	5.0	4.5	4.4	(14)	
(15) Precipitation	PrecAvg	103.2	5.4	2.9	3.2	4.6	8.5	11.1	10.3	10.4	11.4	14.1	11.3	9.3	(15)	
	PrecMax	156.5	27.7	10.9	16.1	12.7	19.4	26.4	25.1	24.5	25.9	26.7	25.1	24.4	(16)	
	PrecMin	70.7	0.0	0.0	0.0	0.4	1.1	2.2	2.4	2.6	3.7	1.2	2.5	0.0	(17)	
	PrecStd	20.1	5.2	2.8	3.2	3.3	4.1	5.7	4.8	5.0	4.2	5.6	4.7	6.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.1	91.0	91.8	93.6	94.8	94.3	93.6	93.6	93.1	92.6	91.7	91.6	(19)	
		MCWB	81.5	81.0	81.4	83.3	84.0	83.1	82.7	82.1	82.5	82.5	82.7	82.2	(20)	
	2%	DB	89.5	89.5	90.6	91.8	93.2	92.8	91.9	91.8	91.5	91.2	90.7	90.0	(21)	
		MCWB	80.5	80.2	80.9	82.4	83.2	82.4	81.8	81.7	81.8	82.1	81.9	81.1	(22)	
	5%	DB	88.1	88.1	89.5	91.2	91.7	91.2	90.9	91.0	90.0	89.7	89.6	89.4	(23)	
		MCWB	79.8	79.7	80.4	82.0	82.4	81.4	81.1	81.3	80.8	81.0	81.1	80.7	(24)	
	10%	DB	87.2	87.2	88.1	89.8	90.3	89.6	89.4	89.5	89.2	88.2	88.1	87.9	(25)	
		MCWB	79.3	79.3	80.0	81.1	81.5	80.5	80.2	80.4	80.3	80.2	80.3	80.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.5	83.3	84.7	85.3	84.4	84.1	84.5	83.9	84.1	84.1	83.5	(27)	
		MCDB	89.6	89.5	90.4	91.8	93.0	92.8	92.3	92.2	91.7	91.3	91.1	90.5	(28)	
	2%	WB	81.3	81.1	81.9	83.6	83.9	83.1	82.8	83.0	82.8	82.7	82.6	82.1	(29)	
		MCDB	88.3	88.0	89.3	91.0	91.9	91.2	90.9	90.7	90.4	90.1	89.8	89.3	(30)	
	5%	WB	80.3	80.2	81.0	82.6	82.9	82.0	81.8	81.8	81.7	81.6	81.6	81.1	(31)	
		MCDB	87.0	86.9	88.2	90.2	90.9	89.9	89.6	89.4	89.2	88.9	88.7	88.2	(32)	
	10%	WB	79.4	79.4	80.2	81.6	81.9	81.0	80.7	80.8	80.7	80.4	80.6	80.1	(33)	
		MCDB	85.6	85.7	87.0	89.0	89.6	88.4	88.2	88.2	87.9	87.4	87.6	86.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	12.8	13.5	14.1	14.3	14.1	14.2	14.1	14.0	13.5	13.5	13.1	(35)	
		MCDBR	14.0	14.0	14.5	15.2	15.5	15.7	15.8	15.6	15.2	14.9	14.8	14.5	(36)	
	5% WB	MCWBR	7.6	7.7	8.1	8.9	9.1	9.3	9.7	9.7	9.2	8.8	8.6	8.1	(37)	
		MCWBR	13.5	13.4	13.9	14.7	15.1	14.9	15.1	14.9	14.8	14.5	14.4	14.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.407	0.414	0.419	0.428	0.419	0.421	0.417	0.450	0.451	0.457	0.418	0.408	(40)	
		taud	2.510	2.466	2.451	2.421	2.463	2.476	2.481	2.364	2.357	2.345	2.470	2.510	(41)	
	Ebn at Noon		289	289	287	279	276	272	274	270	274	273	284	287	(42)	
		Edh at Noon	35	37	37	38	36	35	35	40	41	41	36	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		1462	1691	1826	1897	1753	1634	1623	1676	1637	1543	1520	1428	(44)	
	RadStd		146	163	149	100	93	84	79	89	107	107	99	132	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.36	N/A	N/A	+1.26	+1.03	+0.97	N/A	N/A	+205	+205
(47) Regional (3 neighbors)		+0.23	N/A	N/A	+0.50	+0.54	+0.45	N/A	N/A	+148	+148

Nomenclature: See separate page